

Python Built-in Modules Handbook

Session 9 - Instructor Notes

Chapter 1 - Math Module

sqrt()

Description: Returns the square root.

Syntax: `math.sqrt(x)`

Example:

```
import math
print(math.sqrt(25))
```

Practice: Find sqrt of 64,81,144.

pow()

Description: Raises a number to a power.

Syntax: `math.pow(x, y)`

Example:

```
print(math.pow(2, 3))
```

Practice: Calculate 3^4 and 5^2 .

factorial()

Description: Returns factorial.

Syntax: `math.factorial(x)`

Example:

```
print(math.factorial(5))
```

Practice: Find 4!,6!,8!.

ceil()

Description: Rounds up.

Syntax: `math.ceil(x)`

Example:

```
print(math.ceil(3.2))
```

Practice: Try 7.1 and 9.8.

floor()

Description: Rounds down.

Syntax: `math.floor(x)`

Example:

```
print(math.floor(3.8))
```

Practice: Try 5.9 and 8.2.

pi

Description: Pi constant.

Syntax: `math.pi`

Example:

```
print(math.pi)
```

Practice: Calculate circle area.

fabs()

Description: Absolute value.

Syntax: `math.fabs(x)`

Example:

```
print(math.fabs(-10))
```

Practice: Try -25.

trunc()

Description: Removes decimal part.

Syntax: `math.trunc(x)`

Example:

```
print(math.trunc(7.9))
```

Practice: Try 15.75.

gcd()

Description: Greatest common divisor.

Syntax: `math.gcd(a, b)`

Example:

```
print(math.gcd(20, 30))
```

Practice: Find GCD of 18 and 24.

lcm()

Description: Least common multiple.

Syntax: `math.lcm(a, b)`

Example:

```
print(math.lcm(4, 6))
```

Practice: Find LCM of 8 and 12.

Chapter 2 - Random Module

randint()

Description: Random integer.

Syntax: `random.randint(a,b)`

Example:

```
import random
print(random.randint(1,10))
```

Practice: Generate 10 random numbers.

randrange()

Description: Random value from range.

Syntax: `random.randrange(start, stop, step)`

Example:

```
print(random.randrange(0,20,2))
```

Practice: Generate random even number.

random()

Description: Random float.

Syntax: `random.random()`

Example:

```
print(random.random())
```

Practice: Run several times.

uniform()

Description: Random decimal.

Syntax: `random.uniform(a,b)`

Example:

```
print(random.uniform(1,5))
```

Practice: Generate decimal between 10 and 20.

choice()

Description: Random item.

Syntax: `random.choice(seq)`

Example:

```
colors=['Red','Blue']
print(random.choice(colors))
```

Practice: Choose random fruit.

choices()

Description: Multiple random items.

Syntax: `random.choices(seq,k=3)`

Example:

```
print(random.choices(colors, k=3))
```

Practice: Generate 5 colors.

sample()

Description: Unique random items.

Syntax: `random.sample(seq, k)`

Example:

```
print(random.sample([1, 2, 3, 4], 2))
```

Practice: Pick 3 students.

shuffle()

Description: Shuffle list.

Syntax: `random.shuffle(seq)`

Example:

```
nums=[1, 2, 3, 4]  
random.shuffle(nums)
```

Practice: Shuffle names.

seed()

Description: Repeat random sequence.

Syntax: `random.seed(5)`

Example:

```
random.seed(5)  
print(random.randint(1, 10))
```

Practice: Observe repeated output.

Summary

Use math for calculations and random for generating random values in games, OTPs, simulations, and utilities.